VIT-AP UNIVERSITY

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-23

Faculty

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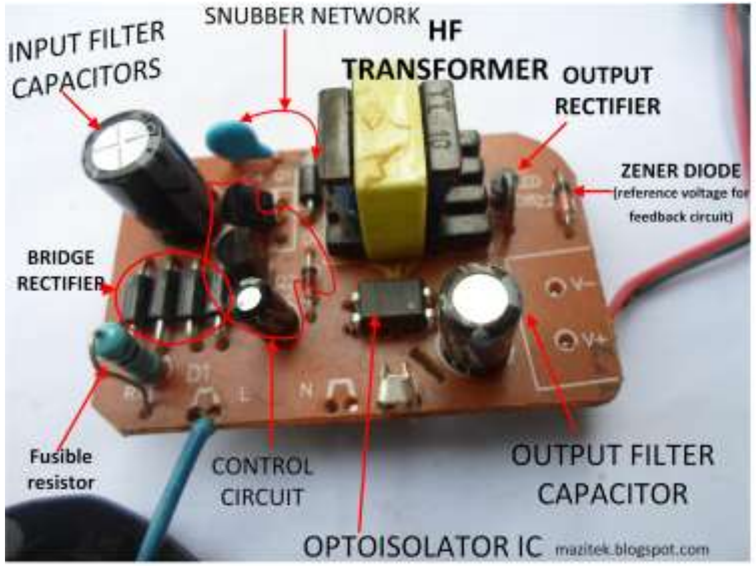
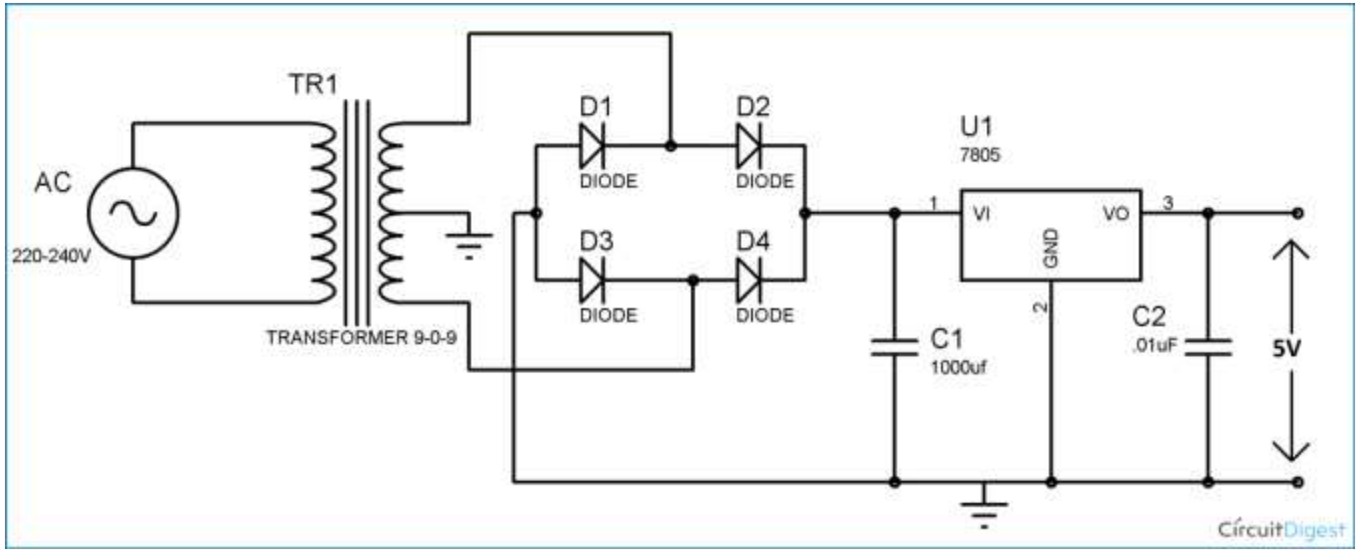
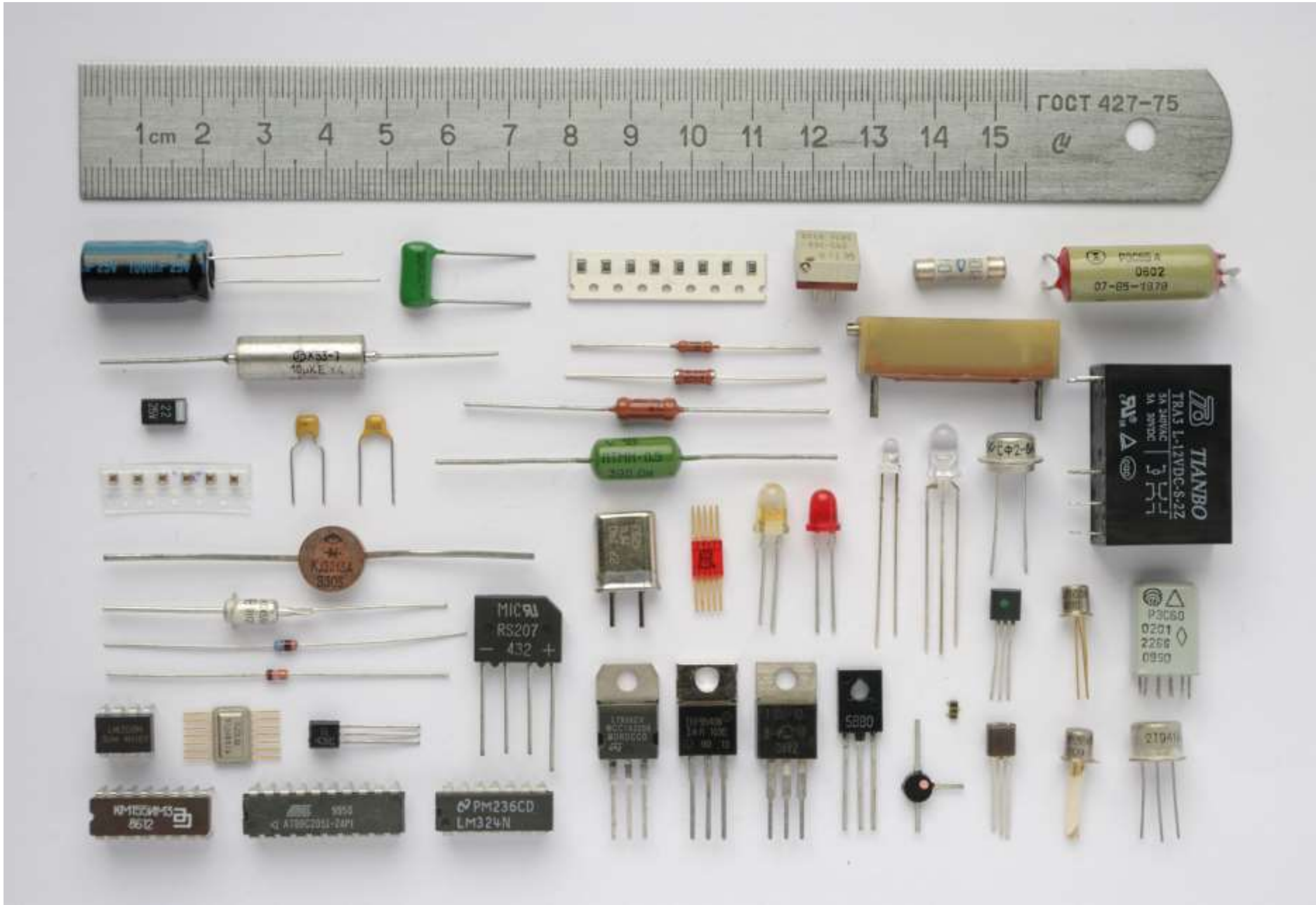
Module-3
Semiconductor Devices and Applications

Mobile Charger





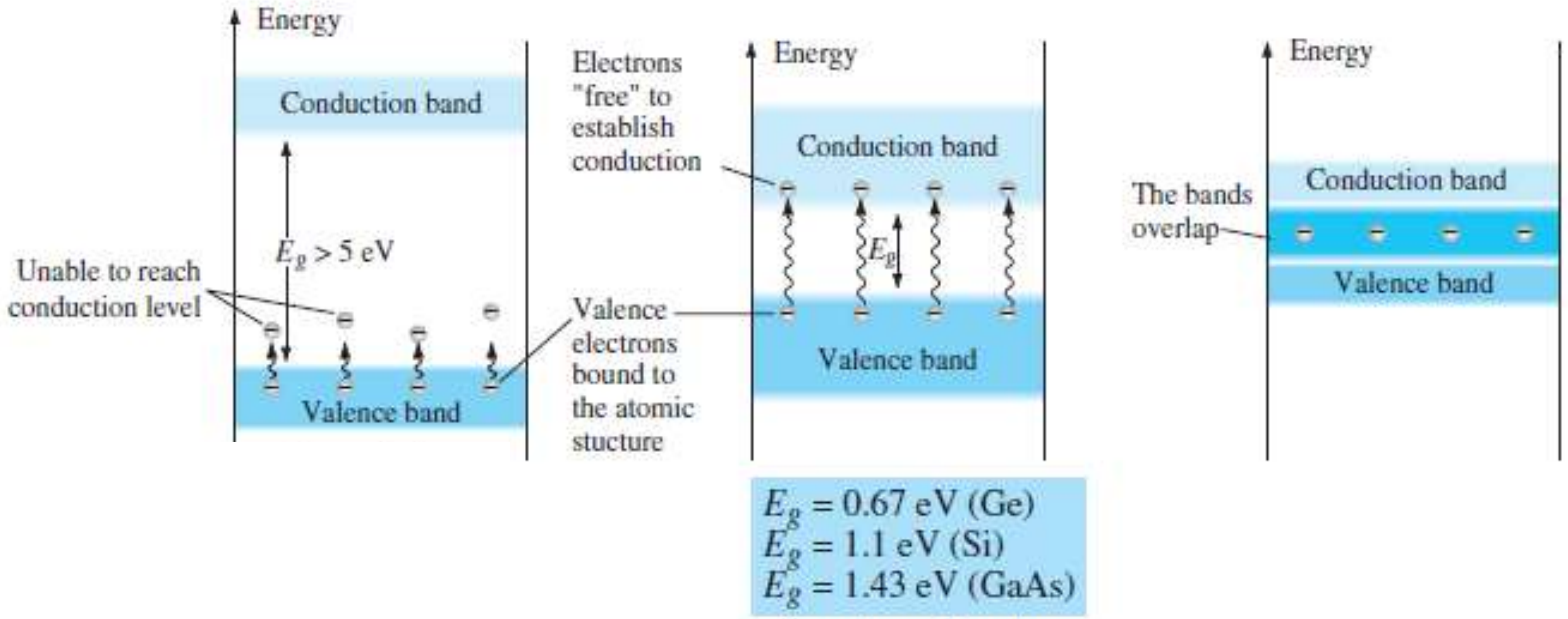

ГОСТ 427-75
1cm 2 3 4 5 6 7 8 9 10 11 12 13 14 15
MIC RS207
432
TIANBO
TRA3 L-12VDC-S-2Z
РЭС60
0201
2266
0950
5880
КМ155ИМ3
8612
PM236CD
LM324N


speaker
monitor
keyboard
tower
television
land line
flatscreen TV
answering machine
share
mouse
smartphone
radio
stereo
walkie-talkie
tablet
webcam
camcorder
earphones
microphone
cell phone
printer
fax machine
CD player
record player
laptop
headset
GPS
remote control
computer
DVD player
digital camera
(USB) flash drive
copier
scanner
timer
MP3 player
watch
calculator

Electronics:

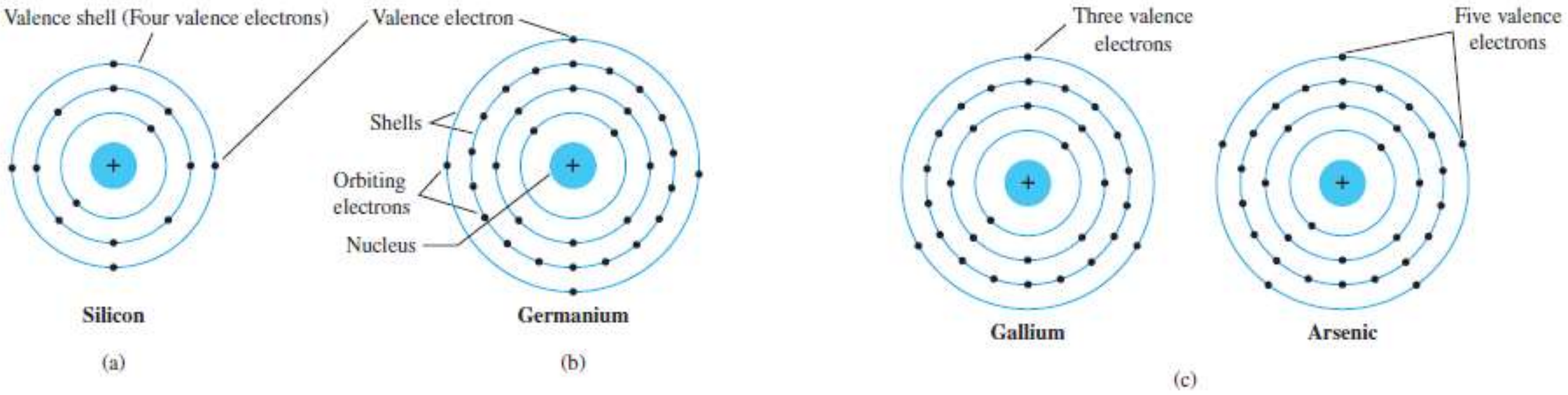
- Electronic devices are components for controlling the flow of electrical currents for the purpose of information processing and system control.
- Prominent examples include transistors and diodes.
- Electronic devices are usually small and can be grouped together into packages called integrated circuits.

Materials-Based on Energy Levels

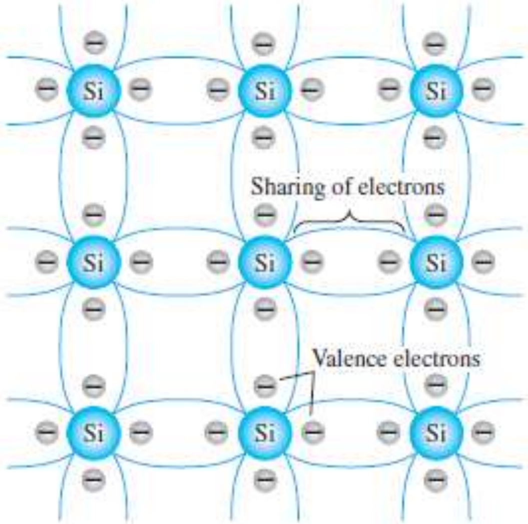


Semiconductor Materials:

- Semiconductors are a special class of elements having a conductivity between that of a good conductor and that of an insulator.
- The three semiconductors used most frequently in the construction of electronic devices are Ge, Si, and GaAs.



Widely used Semiconductor:

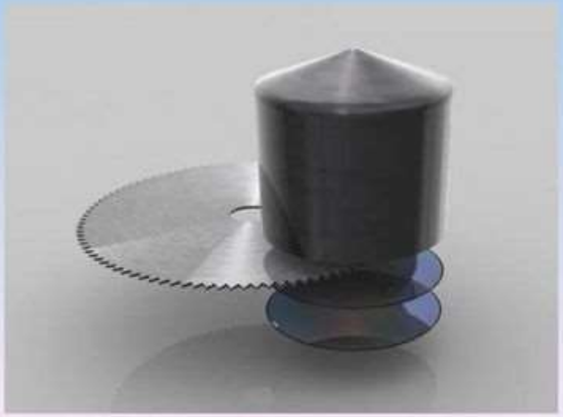


From Sand to Silicon Wafers

Sand, made up of 25% silicon, is the starting point

Purification and heating to make a pure silicon crystal.

100kg of 99.9999% pure silicon crystal



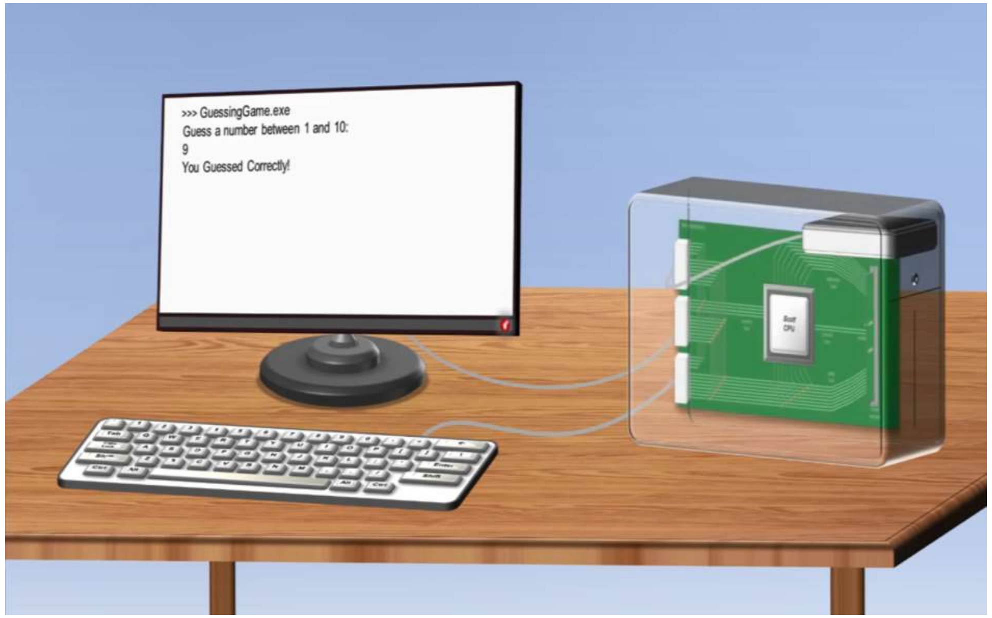
Slicing the crystal into 300mm wafers for use in photolitho-graphy.

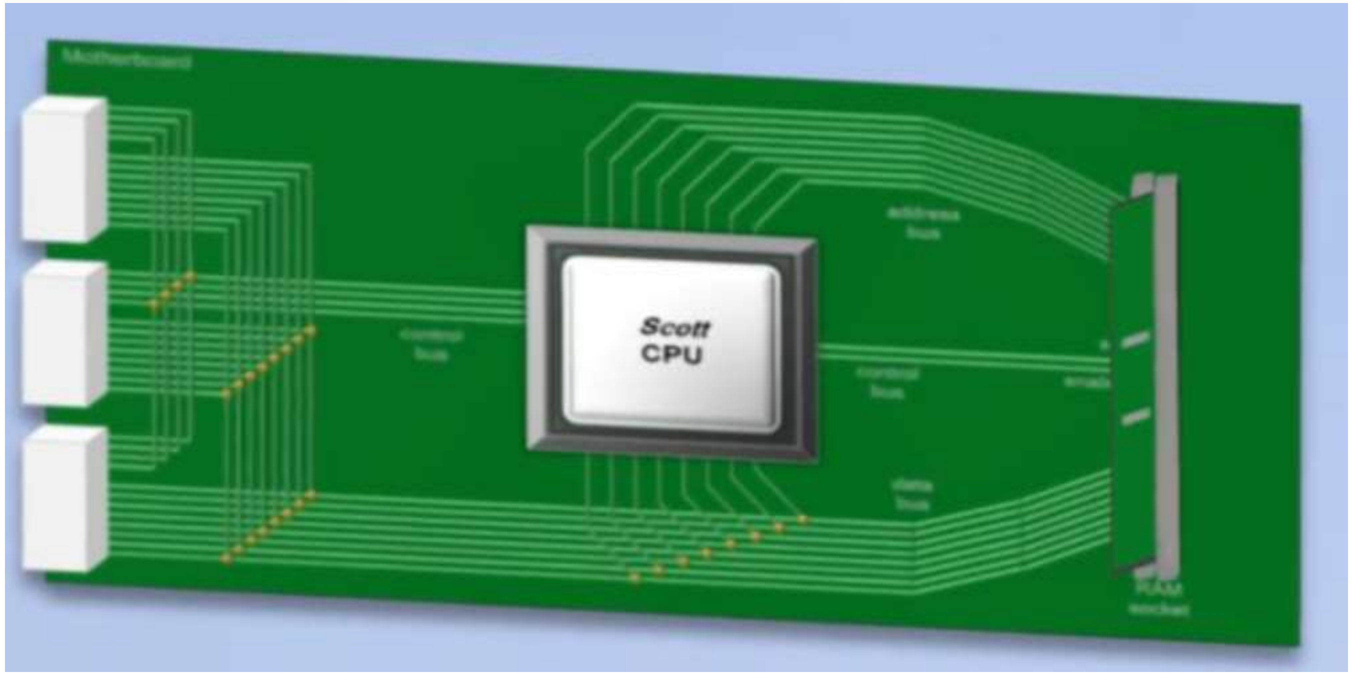
Semiconductor

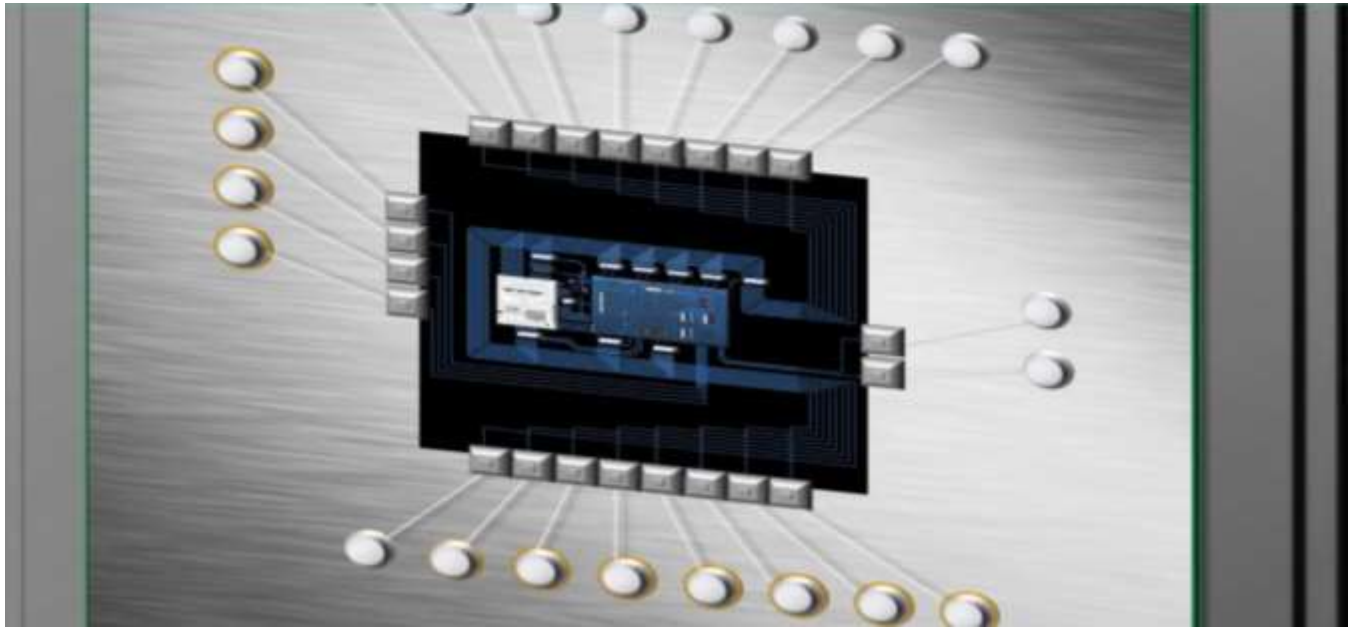
- EXTRINSIC
  - n-type
  - p-type
- INTRINSIC

What's inside a CPU.....??

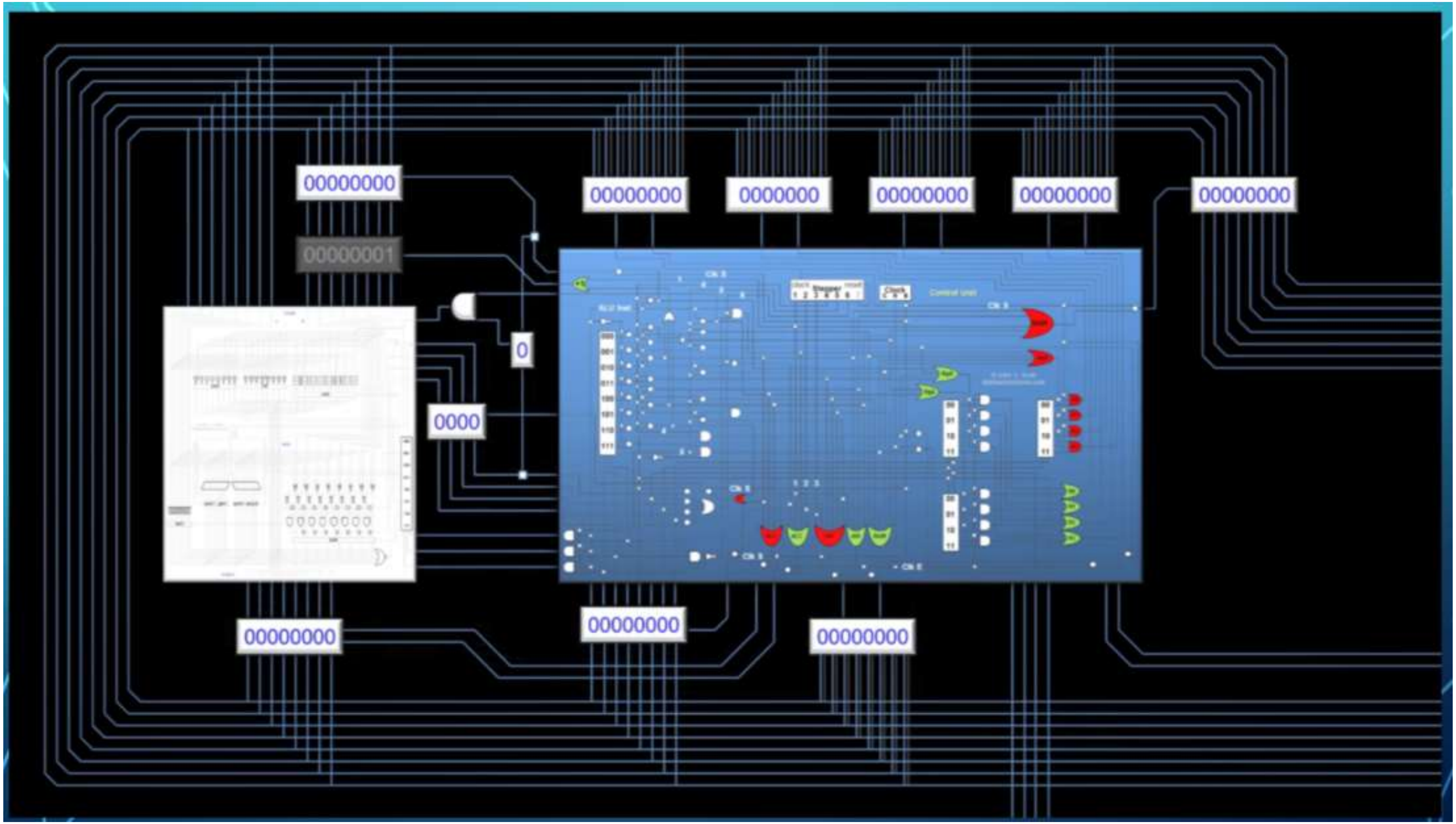
INSIDE PROCESSOR?

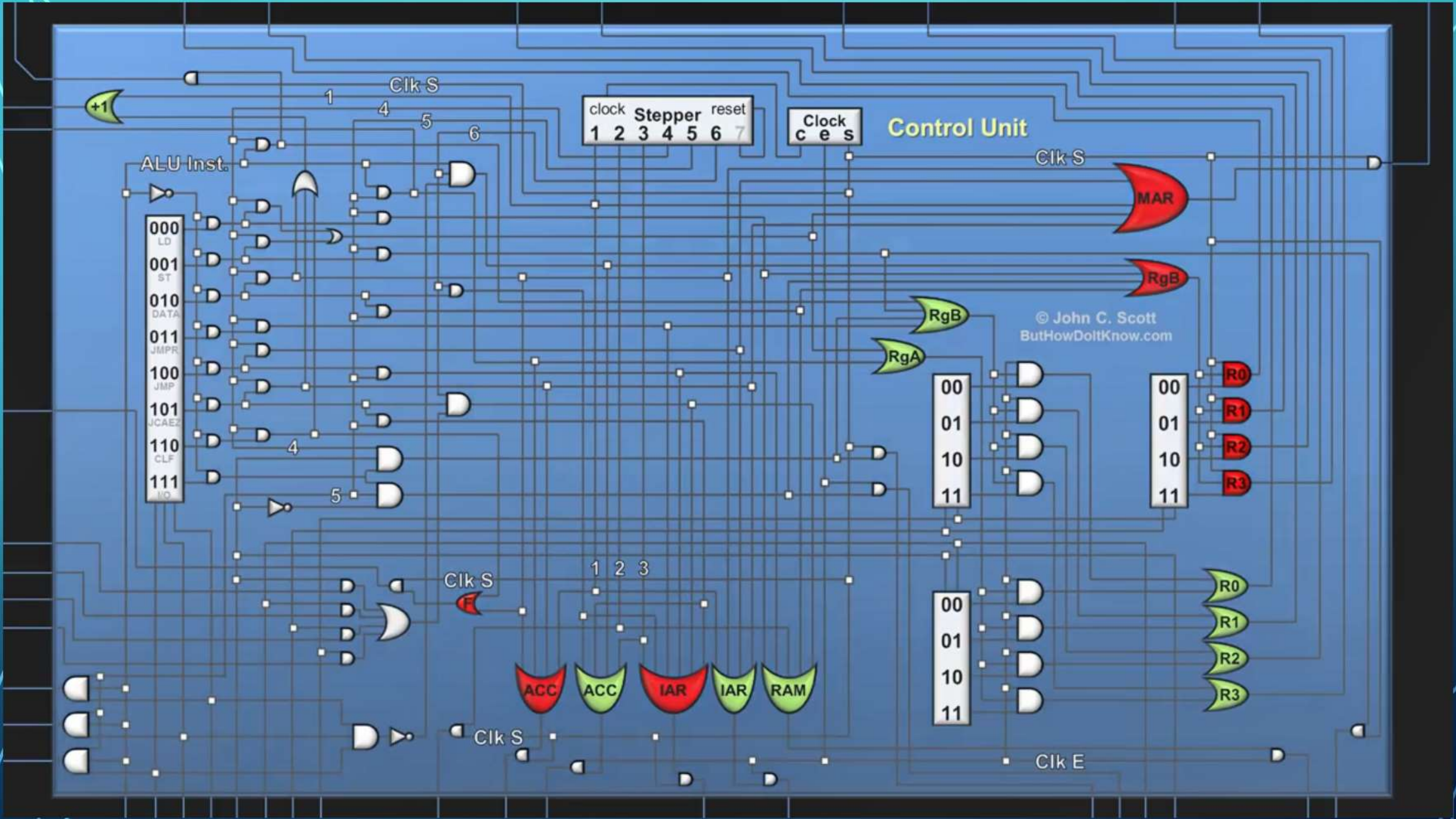






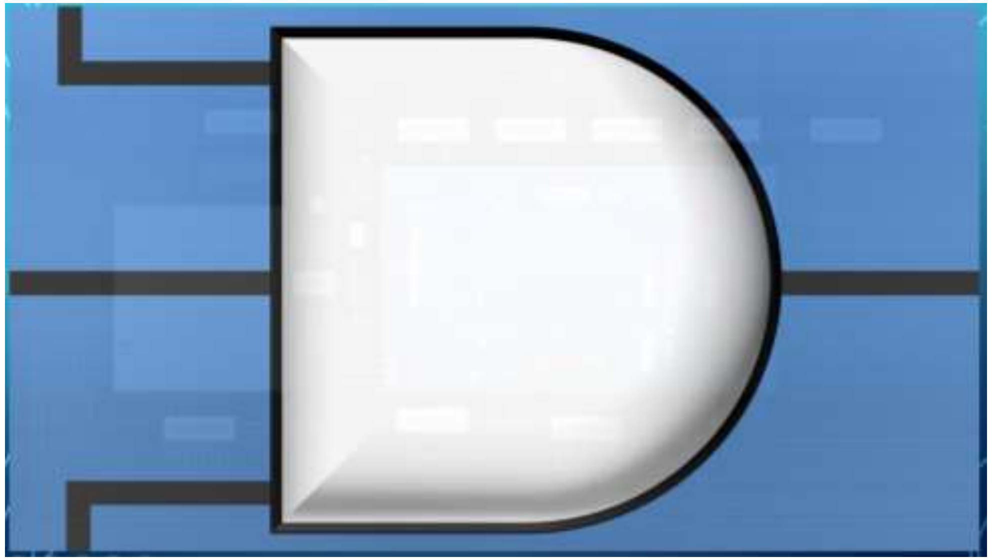
INSIDE PROCESSOR

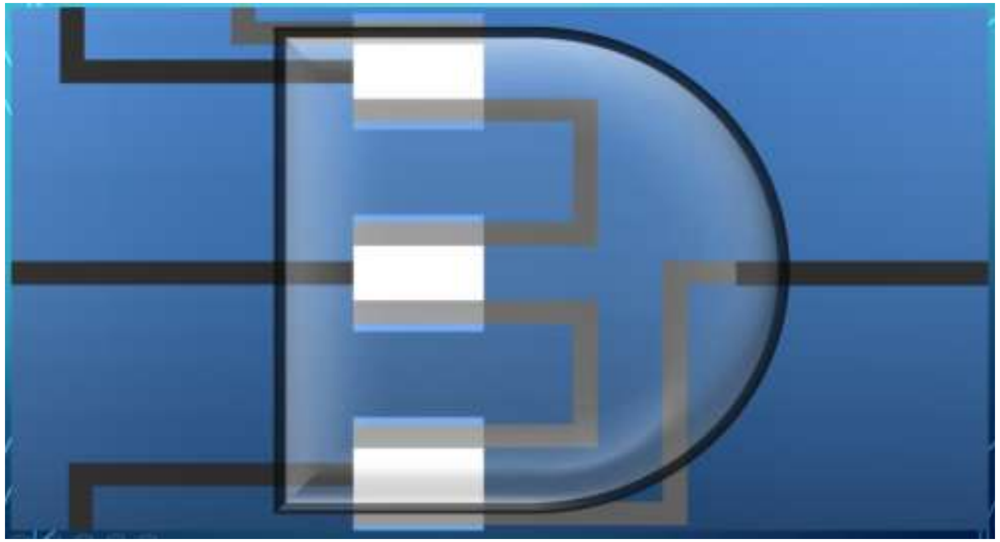


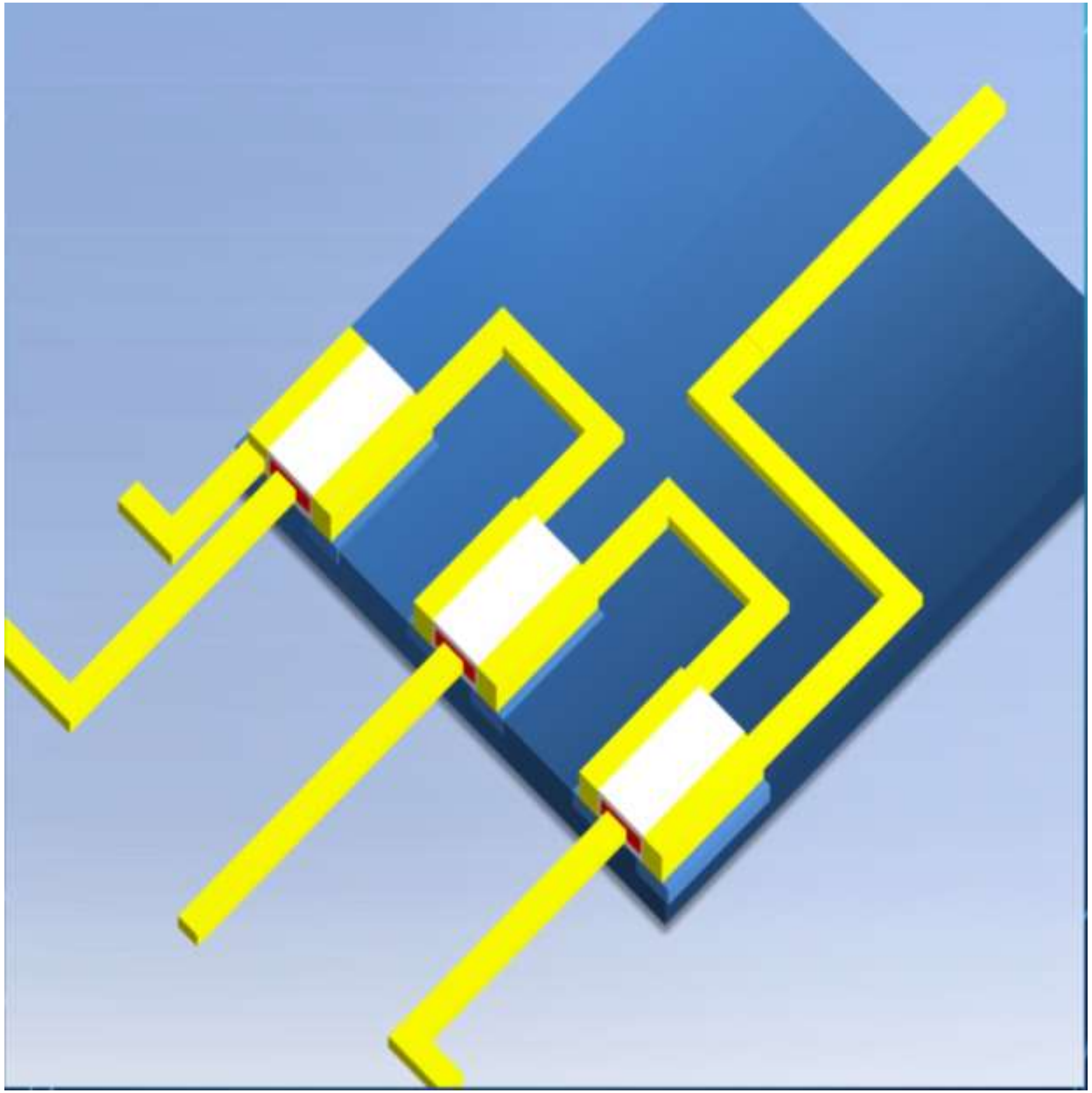


Control Unit
Clk S
+1
1
4
5
6
ALU Inst.
000 LD
001 ST
010 DATA
011 JMPR
100 JMP
101 JCAEZ
110 CLF
111 I/O
clock Stepper reset
1 2 3 4 5 6 7
Clock
c e s
Clk S
MAR
RgB
RgB
RgA
© John C. Scott
ButHowDoItKnow.com
00
01
10
11
R0
R1
R2
R3
4
5
Clk S
F
1 2 3
ACC
ACC
IAR
IAR
RAM
00
01
10
11
R0
R1
R2
R3
Clk S
Clk E

AND GATE

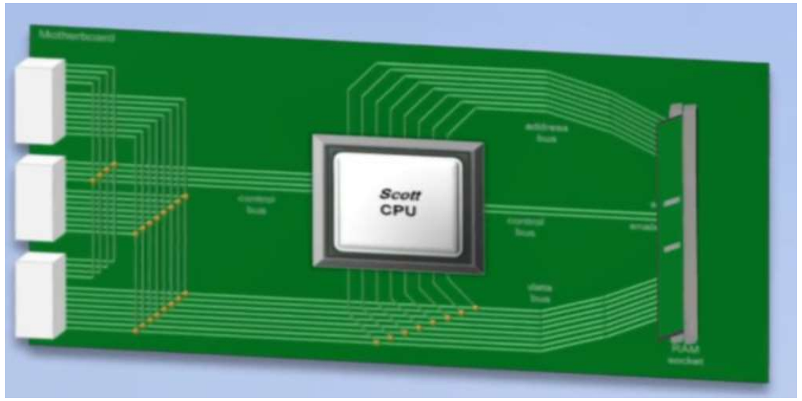




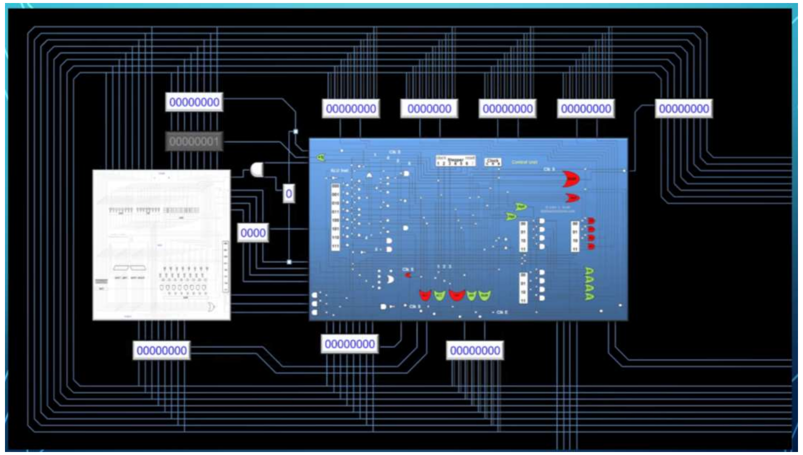


What’s inside a CPU…..??

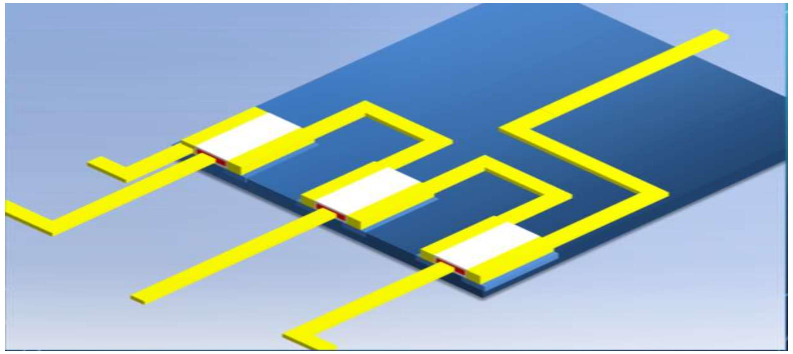
PROCESSOR



LOGIC GATES



TRANSISTOR
S



Processors Manufacturing Companies ????

Intel The Making of a Chip with 22nm_3D Transistors: i5 Processor


intel inside
CORE i5
DESKTOP
INTEL CORE i5 PROCESSOR
intel

Electronic Component: Fabrication Steps